

Urinary System

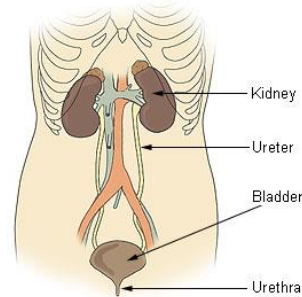
Urinary system consists of set of organs that produce urine in human being.

Party of Urinary System

There are following parts / organs as:

- 1- Kidney
- 2- Ureter
- 3- Bladder
- 4- Urethra

Components of the Urinary System



Functions:

Kidney form urine while the outer parts like ureters, bladder and urethra neither form urine nor change its (urine) composition. They are nearly structure that transport urine from the kidneys to outside the body.

Kidney

Kidneys are a pair of excretory organ situated on the posterior abdominal wall on each side of the vertebral column behind the peritoneum (cavity stomach). The kidneys are mainly located in the lumbar region and also approach the Epigastrium, Hypochondriac and umbilical region. It is reddish brown in color. Each kidney is bean shaped.

Size of kidneys:

12 cm long

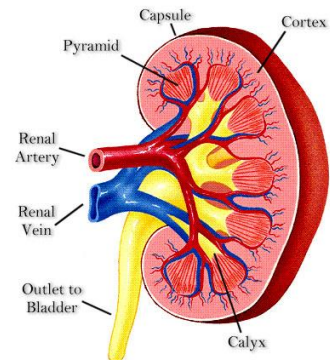
6 cm broad

3 cm thick

The weight of each kidney is 150g in males and 135g in females.

Function of kidney:

Kidney performed many functions:



- i- Maintaining overall fluid balance.
- ii- Regulating and filtering minerals from blood.
- iii- Filtering waste materials from food, medications, and toxic substances.
- iv- Creating hormones that help produce red blood cells, promote bone health, and regulate blood pressure.

Nephron

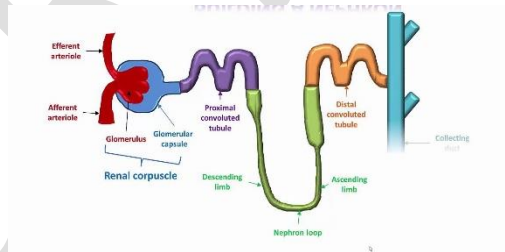
Nephron is the structural and functional unit of kidney which is capable of forming urine.

- ⇒ Each kidney contains millions of nephrons.
- ⇒ Total length of a nephron including collecting duct is about 45-65 mm.

Types of Nephrons:

There are two types of nephrons as:

- 1- Cortical Nephron
- 2- Juxta medullary Nephron



Cortical Nephron:

These are nephrons whose glomeruli lie close to the surface of kidney. These nephrons comprise about 85% of the nephrons in the kidney and have glomeruli located in the Renal Cortex.

Juxta medullary Nephron:

These are nephron whose glomeruli lie deep in the renal cortex near the medulla.

Function:

These Juxta medullary nephrons are specifically instrumental in the production of concentrated urine.

Parts of Nephrons:

There are following parts of nephron:

- i- Glomerulus
- ii- Renal tubules

- iii- Bowman's capsule
- iv- Loop of Henle (Ascending and Descending limb)
- v- Proximal Convolute Tubule
- vi- Distal Convolute Tubule
- vii- Collecting Duct => Ducts of Bellini

Bowman's Capsule:

In each nephron inner end forms a cup-shaped swelling, called Bowman's Capsule.

Glomerulus:

Bowman's capsule is around a ball of capillaries called glomerulus. Its walls are porous. Glomerulus circulates blood through capsule as it arrives through **Afferent Arterioles** and leaves the capsule by **Efferent Arterioles**.

Peritubular Capillaries:

The blood vessel sub-divides again into another network of capillaries called Peritubular capillaries.

⇒ The peritubular capillaries intermingle with proximal and distal tubules of the nephron.

⇒ **Vasa recta:**

In juxta medullary nephron additional capillaries extend down to form a loop of vessels, vasa recta.

⇒ Bowman's capsule continues as extensively convoluted Proximal tubule, loop of Henle, Distal tubule which empties into collecting tubules.

Function of Nephron

The basic function of nephron is to clear and clean the blood plasma of unwanted substances. This occurs in three steps:

- 1- Filtration
- 2- Reabsorption
- 3- Secretion

Filtration:

Blood passing through glomerulus is filtered into Bowman's capsule. It is specifically filtered here, unlike at the other parts of the vessels, because glomerulus walls are porous, and the fraction of the blood pressure reaching here provides the filtration pressure. The filtrate appearing in Bowman's capsule is called as Glomerular filtrate, which contains numerous useful substances such as glucose, amino acid, salt etc. in aqueous solution.

Selective Reabsorption:

The tubular epithelium also secretes substances into the lumen, this secretion is very selective and is mainly of hydrogen ions to balance pH value of the filtrate passing through the tubules.

Urine

This is an amber colored fluid of slight acidic reaction excreted by kidney.

Volume:

The volume of urine excreted in a normal adult varies between 1000-2000 ml/day.

Color:

Pale yellow/amber, depends upon the pigment called Urochrome.

pH:

mean: 6.0, range (4.6-8.0).

Chemical nature:

Acidic

Composition of urine

Organic Substance:

- i- Urea
- ii- Uric acid
- iii- Creatinine
- iv- Ammonia
- v- Enzymes
- vi- Hormones

Inorganic Substance:

- i- Sodium
- ii- Chloride
- iii- Phosphate
- iv- Potassium
- v- Calcium
- vi- Iodine

Abnormal Constituent of Urine:

- i- Protein: Albumin globulin
- ii- Sugar: Glucose, lactose, fructose etc.
- iii- Blood: WBC's, RBC's
- iv- Ketone bodies
- v- Indicant
- vi- Pigments: Bilirubin, melanin
- vii- Casts
- viii- Renal calculi
- ix- Pus
- x- Microbes

Formation of urine:

Fluid is filtered from glomerular capillary into bowman's capsule; this fluid is called "glomerular filtrate".

Basic transport mechanisms through tubular membrane:

There are two transport mechanisms:

- 1- Active transport
- 2- Passive transport

Active transport:

Active transport is the movement of molecules across a membrane from a region of their lower concentration to a region of higher concentration against the concentration gradient. It required cellular energy to active this movements.

- i- Primary active absorptive transport: e.g., Na^+ and Ca^{++}
- ii- Secondary active absorptive transport: glucose and amino acid absorption.
- iii- Secondary active secretory transport

Passive transport:

Passive transport is a movement of ions and other atomic or molecular substances across cell membrane without need of energy input. Unlike active transport.

- i- Osmosis e.g., water
- ii- Diffusion e.g., ions and urea

Formation of dilute / concentration Urine

The kidneys have an ability to alter the composition of urine in response to the body's daily needs. Therefore, the mainly osmolality of the body fluids. When it is necessary to conserve body water, the kidneys excrete urine with high solute concentration. When it is necessary to rid the body of excess water, the kidneys excrete urine with a dilute solute concentration.

Principle Regulator:

The principal regulator of urine composition is ADH. In the absence of ADH the kidney excretes a large volume of dilute urine. When ADH is present in high concentrations, the kidney excretes a small volume of concentration urine.

⇒ ADH released from posterior pituitary lobe act to active transport water from filtrate in collecting tubules back to kidney.

Components of concentrating and diluting system:

The formation of urine that is dilute (hypo-osmotic to plasma) is achieved by the counter current system of nephron. This system consists of a descending limb of the loop of Henle. Thin and thick segments of the ascending limb of the loop of Henle.

Medulla interstitium

Distal convoluted tubule

Collecting duct

Vasa recta, which are the vascular elements of the juxta medullary nephrons.

Summary for dilute urine formation:

- i- Decreased osmolality of ECF
- ii- Decreased secretion of ADH

- iii- Decreased reabsorption of water in distal tubule, cortical collecting tubule and in collecting duct
- iv- More water is excrete in urine
- v- Dilute urine

Summary of concentration urine formation:

- i- Increased osmolality of ECF
- ii- Increased secretion of ADH
- iii- Increased reabsorption of water in late distal tubule, cortical collecting tubules and in collecting duct
- iv- Less water is excreted in urine
- v- Concentration urine